

Pentaerythritol

Pentaerythritol is an organic compound with the formula $\text{C}(\text{CH}_2\text{OH})_4$. The molecular structure can be described as a neopentane with one hydrogen atom in each methyl group replaced by a hydroxyl ($-\text{OH}$) group. It is therefore a polyol, specifically a tetrol.

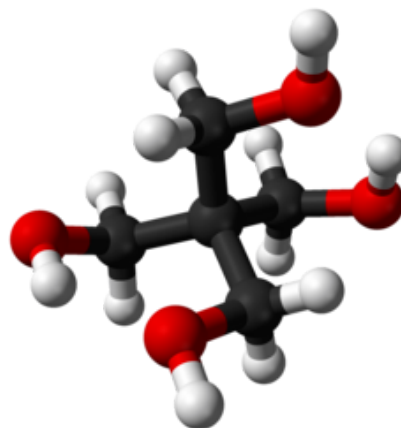
Pentaerythritol is a white solid. It is a building block for the synthesis and production of explosives, plastics, paints, appliances, cosmetics, and many other commercial products.

The word pentaerythritol is a blend of *penta-* in reference to its five carbon atoms and *erythritol*, which also possesses 4 alcohol groups.

Synthesis

Pentaerythritol was first reported in 1891 by German chemist Bernhard Tollens and his student P. Wigand.^[5] It may be prepared via a base-catalyzed multiple-addition reaction between acetaldehyde and 3 equivalents of formaldehyde to give pentaerythrose (CAS: 3818-32-4), followed by a Cannizzaro reaction with a fourth equivalent of formaldehyde to give the final product plus formate ion.^[6]

Pentaerythritol



Names

Preferred IUPAC name

2,2-Bis(hydroxymethyl)propane-1,3-diol^[1]

Other names

2,2-Bis(hydroxymethyl)-1,3-propanediol

Pentaerythritol^[1]

Hercules P 6

Monopentaerythritol

Tetramethylolmethane

THME

PETP

Pentaerythrite

Pentek

Hercules Aqualon improved technical PE-200

Identifiers

CAS Number

115-77-5 (https://commonchemistry.cas.org/detail?cas_rn=115-77-5)[✓]

3D model (JSmol)

Interactive image (<https://chemapps.stolaf.edu/jmol/jmol.p>

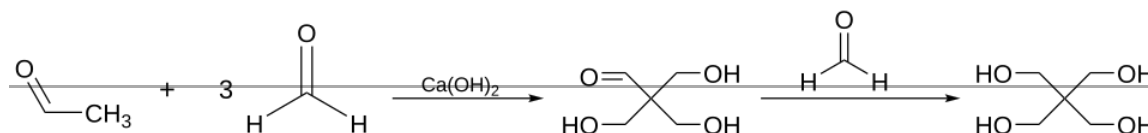
	hp?model=OCC%28CO%29%28CO%29CO)
ChEBI	CHEBI:134760 (https://www.ebi.ac.uk/chebi/searchId.do?chebiId=134760)
ChEMBL	ChEMBL3186112 (https://www.ebi.ac.uk/chembl/exlore/compound/ChEMBL3186112)
ChemSpider	7984 (https://www.chemspider.com/Chemical-Structure.7984.html) ✓
DrugBank	DB13526 (https://www.drugbank.ca/drugs/DB13526)
ECHA InfoCard	100.003.732 (https://echa.europa.eu/substance-information/-/substanceinfo/100.003.732)
EC Number	204-104-9
KEGG	D08331 (https://www.kegg.jp/entry/D08331)
PubChem CID	8285 (https://pubchem.ncbi.nlm.nih.gov/compound/8285)
RTECS number	RZ2490000
UNII	SU420W1S6N (https://precision.fda.gov/uniisearch/srs/unii/SU420W1S6N) ✓
CompTox Dashboard (EPA)	DTXSID2026943 (https://comptox.epa.gov/dashboard/chemical/details/DTXSID2026943)
InChI	
InChI=1S/C5H12O4/c6-1-5(2-7,3-8)4-9/h6-9H,1-4H2 ✗	
Key: WXZMFSXDPGVJJK-UHFFFAOYSA-N ✗	
InChI=1/C5H12O4/c6-1-5(2-7,3-8)4-9/h6-9H,1-4H2	
Key: WXZMFSXDPGVJJK-UHFFFAOYAH	
SMILES	
OCC(CO)(CO)CO	
Properties	
Chemical formula	C(CH ₂ OH) ₄

<u>Molar mass</u>	136.147 g·mol ⁻¹
<u>Appearance</u>	white crystalline solid
<u>Odor</u>	Odorless
<u>Density</u>	1.396 g/cm ³
<u>Melting point</u>	260.5 °C (500.9 °F; 533.6 K)
<u>Boiling point</u>	276 °C (529 °F; 549 K) at 30 mmHg
<u>Solubility in water</u>	38.46 g/L (0°C) 47.62 g/L (10°C) 52.60 g/L (15°C) 56.60 g/L (20°C) 74.07 g/L (30°C) 115.0 g/L (40°C) 180.3 g/L (60°C) 285.7 g/L (80°C) 500.0 g/L (100°C) ^[2]
<u>Solubility</u>	TBuOH, 15g/L (60°C) ^[3] DMSO, 20g/L (25°C) Slightly soluble in: <u>methanol</u> , <u>ethanol</u> , <u>glycerol</u> , <u>ethylene glycol</u> , <u>formamide</u> ; insoluble in: <u>acetone</u> , <u>toluene</u> , <u>heptane</u> , <u>diethyl ether</u> , <u>dichloromethane</u>
<u>Vapor pressure</u>	0.00000008 mmHg (20°C) ^[4]
Hazards	
<u>Flash point</u>	200.1 °C (392.2 °F; 473.2 K)
NIOSH (US health exposure limits):	
<u>PEL</u> (Permissible)	TWA 15 mg/m ³ (total) TWA 5 mg/m ³ (resp) ^[4]
<u>REL</u> (Recommended)	TWA 10 mg/m ³ (total) TWA 5 mg/m ³ (resp) ^[4]
<u>IDLH</u> (Immediate danger)	N.D. ^[4]
Related compounds	
<u>Related compounds</u>	<u>Neopentane</u> <u>Neopentyl alcohol</u> <u>Neopentyl glycol</u> <u>Trimethylolethane</u>

Except where otherwise noted, data are given for materials in their standard state (at 25 °C [77 °F], 100 kPa).

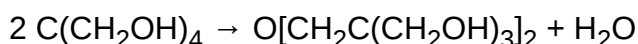
✗ verify (what is ✓ ✗ ?)

Infobox references



Uses

Pentaerythritol is a versatile building block for the preparation of many compounds,^[7] particularly polyfunctionalized derivatives. applications include alkyd resins, varnishes, polyvinyl chloride stabilizers, tall oil esters, antioxidants (e.g. Anox 20). Such derivatives are found in plastics, paints, cosmetics, and many other products. Relevant to resins, pentaerythritol is a precursor to other polyol, such as dipentaerythritol.^[8]



Esters of pentaerythritol are biodegradable,^{[9][10]} and they are used as transformer oils.^[11] Due to a very high flash point they also find some use in lubricating gas turbines.^[12]

Ester derivatives

Pentaerythritol is a precursor to esters of the type $\text{C}(\text{CH}_2\text{OX})_4$. Such derivatives are pentaerythritol tetranitrate (PETN), a vasodilator and explosive, the trinitrate derivative pentritinol (Petrin), the tetraacetate normosterol (PAG), and the polymer cross-linking agents pentaerythritol tetraacrylate and pentaerythritol tetrakis(3-mercaptopropionate).^{[13][14]}

A linear polymer which can be described as a (spiro) orthocarbonate ester of pentaerythritol, whose formula could be written as $[(-\text{CH}_2)_2\text{C}(\text{CH}_2-)_2(-\text{O})_2\text{C}(\text{O}-)_2]_n$, was synthesized in 2002.^[15]

Fire retardants

Pentaerythritol is used as a fire retardant, such as in plastics and intumescent paints and coatings. It releases water upon heating and leaves a deposit of thermally insulating char.^[16]

See also

- 1,6-Hexanediol
- Neopentyl glycol
- Trimethylolethane
- Trimethylolpropane

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